Huxley-Gödel Machine: Select lineages, not just high-scoring coding agents

Wenyi Wang · Piotr Piękos · Li Nanbo · Firas Laakom · Yimeng Chen
Mateusz Ostaszewski · Mingchen Zhuge · Jürgen Schmidhuber · KAUST · arXiv 2510.21614v3

HarnessRSI 63 ★★

evidence: retained

Paper

Code

1 What rewrites what?

Archive of agent repositories
A coding agent edits itself into a child.
Code is retained; no weight training described.

Choose expansion or evaluation
Widening rule uses task count and tree size.
All experiments use asynchronous HGM.

Action A: expand a promising clade
Thompson-sample using subtree successes and failures, including the parent itself.
Self-modify; add child before evaluation.

OR

Action B: evaluate on one task
Separately sample using its own outcomes.
Pass/fail updates this node and ancestors.
No shown child-must-beat-parent gate.

Budget ends → select final artifact
Return the stated best-belief agent.
Posterior-percentile notation is unresolved (see controls); no implementation inferred.

Method: §§2–3, Algorithm 1. The outer search policy and benchmark utility are specified externally; protected files are not listed.

Limits: finite budget; one-hour execution cap in comparisons, five hours at full scale. Not a safety guarantee.

2 Better selected agents, less CPU time

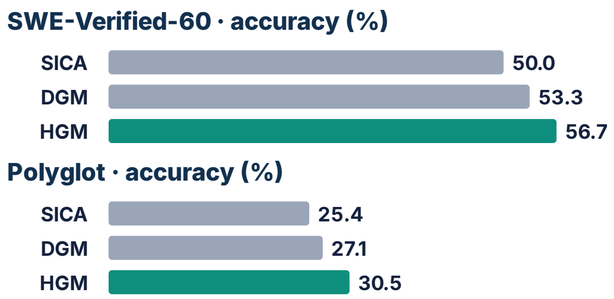


Table 2 · up to 800 task evaluations; optimization/selection tasks, not held-out tests. Shared ancestor: 40% SWE, 20.3% Polyglot.
Allocated CPU-hours: HGM 517 / 347; DGM 1231 / 2385 (SWE / Polyglot). SICA: SWE context errors after 360 evaluations; Polyglot 572 h.

4 Where the edge becomes thin

Lite · filtered / standard	HGM	SWE-agent
GPT-5-mini (Table 3)	40.1 / 49.0	39.6 / 47.6
GPT-5 transfer (Table 4)	47.8 / 57	48.3 / 56.7

Accuracy %. Table 3: local GPT-5-mini baseline. Table 4: SWE-agent + Claude 4 Sonnet; not matched-backbone.

- Filtered GPT-5: **one task behind** SWE-agent. GPT-5-mini edge is small; no CIs.
- "Human-level": human-engineered agents, not human coders; checked Lite LB (24 Oct 2025).

3 Controls and unresolved details

- **Same start, different policies.** DGM/SICA share the ancestor, models and evaluation ceiling. SICA sees all history; DGM evaluates progressively. Neither gets HGM's clade/async policy.
- **No isolated ablation:** CMP and async change together. No seed/run counts, SDs or CIs in Tables 1–4; do not infer single-run.
- **Diagnostic:** weighted CMP correlation 0.778 vs DGM 0.285 (SWE); 0.626 vs 0.383 (Polyglot). Root + winning branch excluded; not a prospective test.
- **Held-out:** Verified → Lite. Filtered Lite removes 93 overlaps from 300 tasks. Polyglot hides private tests/results; no broad contamination audit.
- **Tuning:** $\alpha = 0.6$, $\varepsilon = 1$, $\tau = B/b$. Selection split and baseline tuning budgets unreported; interaction-length cap, decoding temperature and worker count not numerically specified.
- **Cost:** ~$5000 across all methods; no full-run HGM-only breakdown. Task-call matching ≠ dollar/token matching.
- **Async detail:** warm start described separately from base Algorithm 1. **Unresolved:** percentile prose vs Beta-CDF rule ($\varepsilon = 1$) leaves final ranking undefined.
- **Text/table:** Polyglot unweighted correlation 0.8783 in prose vs 0.873 in Table 1; table used, cause unresolved.

Models (modify / evaluate): SWE GPT-5 / GPT-5-mini; Polyglot Qwen3-Coder-480B / 30B. Both domains gain; no per-repository breakdown.

Takeaway

Useful finite-budget search over self-modifying code: value a lineage, not only its parent's score. Filtered Lite supports retained improvement, but the small external margins, missing uncertainty and bundled policy changes do not prove an optimal or endlessly accelerating improver.

Our read · HarnessRSI

Score 63 · ★★ · evidence: retained

Reports: transfer, partial budget match, CPU cost, narrow leakage guards.
Missing: uncertainty, isolated ablations, broad contamination audit.